

Marko Kosunen

associate professor

基本信息

姓名: Marko Kosunen

所属院校: 阿尔托大学

地区: Finland

年龄范围: 40岁至50岁

学科领域

通信工程

微电子科学与工程

集成电路设计与集成系统

研究领域

无线通信与智能天线

移动通信与5G/6G

射频集成电路

模拟与射频IC设计

通信系统仿真与优化

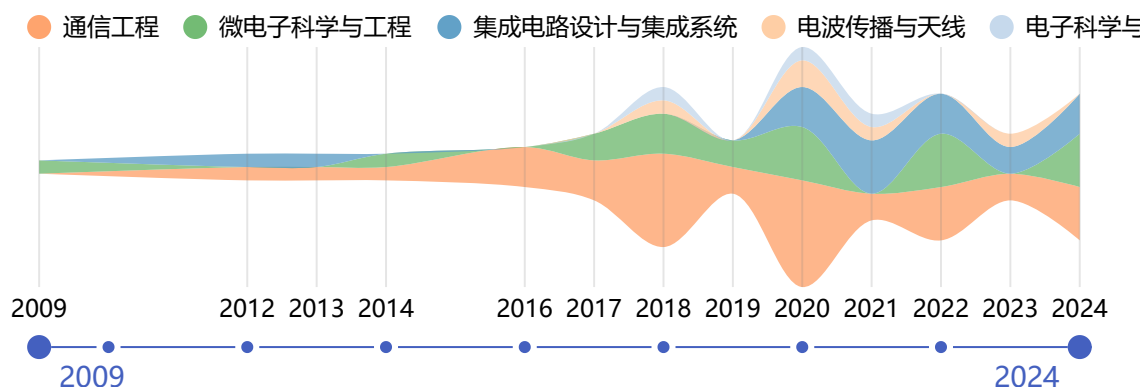
通信系统与网络 (通信工程)

模拟与数模混合电路

个人简介

Marko Kosunen is an Associate Professor at Aalto University's Department of Electronics and Nanoengineering. He holds M.Sc, L.Sc, and D.Sc (with honors) degrees from Helsinki University of Technology. From 2017 - 2019, he visited UC Berkeley on a Marie Skłodowska - Curie grant. He co - chairs Microelectronics Finland. Kosunen has over a hundred journal and conference papers and several patents. His research interests include circuit design, data converters, transceiver circuits, and RISC - V microprocessors. He's involved in projects like EffiCores and TIRESYAS. Contact: +358 50 3751711, Maarintie 8 02150 Espoo, <https://orcid.org/0000-0002-2723-1859>.

近年发表的论文



Variable Time-Step 2.5-15 GHz Phase Modulator with Pre-Distortion for Outphasing Transmitters

2024 22ND IEEE INTERREGIONAL NEWCAS CONFERENCE, NEWCAS 2024

Cheung, Tze Hin; Ghosh, Agnimesh; Spelman, Andrei; Boopathy, Dhanashree; Unnikrishriant, Vishnu; Antfilat, Lauri; Valkamat, Mikko; Kosunen, Marko; Stadius, Kari; Ryyntinen, Jussi

被引用次数: 0

2024

Analysis and Design of Constant-Slope Voltage-to-Time Converters

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Porrasmaa, Santeri; Jarvinen, Okko; Kempfi, Ilia; Stadius, Kari; Kosunen, Marko; Ryyntinen, Jussi

被引用次数: 0

2024

Processor configurable sample rate converter for RISC-V and 5G communication

2024 22ND IEEE INTERREGIONAL NEWCAS CONFERENCE, NEWCAS 2024

Simola, Otto; Spelman, Andrei; Korsman, Aleks; Kosunen, Marko; Ryyntinen, Jussi

被引用次数: 0

2024

An 18-28 GHz dual-mode down-converter IC for 5G applications

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Naghavi, Saeed; Ryyntinen, Kaisa; Zahra, Mahwish; Korsman, Aleks; Stadius, Kari; Kosunen, Marko; Unnikrishnan, Vishnu; Anttila, Lauri; Valkama, Mikko; Ryyntinen, Jussi

被引用次数: 0

2024

A 22-nm All-Digital Time-Domain Neural Network Accelerator for Precision In-Sensor Processing

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Mohey, Ahmed M.; Leslin, Jelin; Singh, Gaurav; Kosunen, Marko; Ryyntinen, Jussi; Andraud, Martin

被引用次数: 0

2024

Reconfigurable Signal Processing and DSP Hardware Generator for 5G and Beyond Transmitters

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ghosh, Agnimesh; Spelman, Andrei; Cheung, Tze Hin; Boopathy, Dhanashree; Stadius, Kari; Gomony, Manil Dev; Valkama, Mikko; Ryyntinen, Jussi; Kosunen, Marko; Unnikrishnan, Vishnu

被引用次数: 1

2024

Multilevel Outphasing With Over-the-Air Combining in Large Antenna Arrays

IEEE TRANSACTIONS ON COMMUNICATIONS

Lampu, Vesa; Xu, Guixian; Brihuega, Alberto; Kosunen, Marko; Unnikrishnan, Vishnu; Ryyanen, Jussi; Fager, Christian; Valkama, Mikko; Anttila, Lauri

被引用次数: 1

2023

Toward All-Digital Time-Domain Neural Network Accelerators for In-Sensor Processing Applications

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Mohey, Ahmed M.; Kosunen, Marko; Ryyanen, Jussi; Andraud, Martin

被引用次数: 0

2023

An Automated EM-Simulation Environment with Parameterized Layout Generation for Microwave Integrated Circuits

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Ryyanen, Kaisa; Lahtinen, Veeti; Porrasmaa, Santeri; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 0

2023

Leveraging frequency agility of an MIMO antenna cluster with a transmitter IC

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Saleem, Ali Raza; Luomaniemi, Rasmus; Lehtovuori, Anu; Stadius, Kari; Kosunen, Marko; Viikari, Ville; Ryyanen, Jussi

被引用次数: 0

2023

A 0.9-Nyquist-Band Digital Timing Mismatch Correction for Time-Interleaved ADCs Achieving Delay Tuning Range of 0.12-Sample-Period

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Kempi, Ilia; Jarvinen, Okko; Kosunen, Marko; Unnikrishnan, Vishnu; Stadius, Kari; Ryyanen, Jussi

被引用次数: 1

2022

Energy-Efficient Cyclic-Coupled Ring Oscillator With Delay-Based Injection Locking

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jarvinen, Okko; Unnikrishnan, Vishnu; Kempf, Ilia; Porrasmaa, Santeri; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 2

2022

True-Time-Delay Receiver IC With Reconfigurable Analog and Digital Beamforming

IEEE ACCESS

Spoof, Kalle; Tenhunen, Miikka; Unnikrishnan, Vishnu; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 3

2022

Design of Cyclic-Coupled Ring Oscillators with Guaranteed Maximal Phase Resolution

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Jarvinen, Okko; Unnikrishnan, Vishnu; Kempf, Ilia; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 0

2022

A 6-20 GHz 400-MHz Modulation-Bandwidth CMOS Transmitter IC

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Saleem, Ali Raza; Naghavi, Saeed; Zahra, Mahwish; Stadius, Kari; Kosunen, Marko; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 2

2022

A 0.8-6 GHz True-Time-Delay Beam-Nulling Receiver Front-End

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Spoof, Kalle; Tenhunen, Miikka; Unnikrishnan, Vishnu; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 0

2022

Reconfigurable Signal Processing and DSP Hardware Generator for 5G Transmitters

2022 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE (NORCAS)

Ghosh, Agnimesh; Spelman, Andrei; Cheung, Tze Hin; Boopathy, Dhanashree; Unnikrishnan, Vishnu; Lampu, Vesa; Xu, Guixian; Anttila, Lauri; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 3

2022

Data Conversion With Subgate-Delay Time Resolution Using Cyclic-Coupled Ring Oscillators

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Unnikrishnan, Vishnu; Jarvinen, Okko; Siddiqui, Waqas; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 5

2021

Quantized Polar Transmitters for Power Efficient Massive MIMO Systems

IEEE WIRELESS COMMUNICATIONS LETTERS

Xu, Guixian; Lampu, Vesa; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko; Anttila, Lauri

被引用次数: 1

2021

Characterization of an Antenna Cluster and Transmitter IC with a Modulated Signal

2021 IEEE MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM (IMS)

Saleem, Ali Raza; Stadius, Kari; Hannula, Jari-Matti; Lehtovuori, Anu; Kosunen, Marko; Viikari, Ville; Ryyanen, Jussi

被引用次数: 0

2021

A 100-750 MS/s 11-Bit Time-to-Digital Converter With Cyclic-Coupled Ring Oscillator

IEEE ACCESS

Jarvinen, Okko; Unnikrishnan, Vishnu; Siddiqui, Waqas; Korhonen, Teuvo; Koli, Kimmo; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 9

2021

A Transmitter IC with Supply Tuning for Frequency-Reconfigurable Antenna Cluster

2021 IEEE INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS (MWSCAS)

Saleem, Ali Raza; Stadius, Kari; Hannula, Jari-Matti; Lehtovuori, Anu; Kosunen, Marko; Viikari, Ville; Ryyanen, Jussi

Passive Intermodulation in Simultaneous Transmit-Receive Systems: Modeling and Digital Cancellation Methods

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Waheed, Muhammad Zeeshan; Korpi, Dani; Anttila, Lauri; Kiayani, Adnan; Kosunen, Marko; Stadius, Kari; Campo, Pablo Pascual; Turunen, Matias; Allen, Markus; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 27

2020

A wideband blocker-resilient direct delta sigma receiver with selective input-impedance matching

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Ul Haq, Faizan; Englund, Mikko; Ostman, Kim B.; Stadius, Kari; Kosunen, Marko; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 0

2020

A Six-Phase Two-Stage Blocker-Tolerant Harmonic-Rejection Receiver

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ul Haq, Faizan; Englund, Mikko; Antonov, Yury; Tenhunen, Miikka; Stadius, Kari; Kosunen, Marko; Ostman, Kim B.; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 4

2020

Time-Based Sensor Interface for Dopamine Detection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Olabode, Olaitan; Kosunen, Marko; Unnikrishnan, Vishnu; Palomaki, Tommi; Laurila, Tomi; Halonen, Kari; Ryyanen, Jussi

被引用次数: 5

2020

True-Time-Delay Beamforming Receiver With RF Re-Sampling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Spoof, Kalle; Unnikrishnan, Vishnu; Zahra, Mahwish; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

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2020

A Class-D Tri-Phasing CMOS Power Amplifier With an Extended Marchand-Balun Power Combiner

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Martelius, Mikko; Stadius, Kari; Lemberg, Jerry; Roverato, Enrico; Nieminen, Tero; Antonov, Yury; Anttila, Lauri; Valkama, Mikko; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 5

2020

A 2-5.5 GHz Beamsteering Receiver IC With 4-Element Vivaldi Antenna Array

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Zahra, Mahwish; Kempf, Ilia; Haarla, Jaakko; Antonov, Yury; Khonsari, Zahra; Miilunpalo, Toni; Ahmed, Nouman; Inkinen, Juha; Unnikrishnan, Vishnu; Lehtovuori, Anu; Viikari, Ville; Anttila, Lauri; Valkama, Mikko; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi

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2020

A 1.5-5-GHz Integrated RF Transmitter Front End for Active Matching of an Antenna Cluster

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Saleem, Ali Raza; Stadius, Kari; Hannula, Jari-Matti; Lehtovuori, Anu; Kosunen, Marko; Viikari, Ville; Ryynanen, Jussi

被引用次数: 9

2020

Resilient flow control for wireless data streaming in inductively coupled medical implants

MICROPROCESSORS AND MICROSYSTEMS

Kempi, Ilia; Ahmed, Nouman; Hammer, Andreas; Olabode, Olaitan; Unnikrishnan, Vishnu; Kosunen, Marko; Ryynanen, Jussi

被引用次数: 9

2020

A 1.5-1.9-GHz All-Digital Tri-Phasing Transmitter With an Integrated Multilevel Class-D Power Amplifier Achieving 100-MHz RF Bandwidth

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lemberg, Jerry; Martelius, Mikko; Roverato, Enrico; Antonov, Yury; Nieminen, Tero; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Kosunen, Marko; Ryynanen, Jussi

被引用次数: 24

2019

Quantization noise upconversion effects in mixer-first direct delta-sigma receivers

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Ul Haq, Faizan; Englund, Mikko; Ostman, Kim B.; Stadius, Kari; Kosunen, Marko; Koli, Kimmo; Ryynanen, Jussi

被引用次数: 0

2019

Tri-Phasing Modulation for Efficient and Wideband Radio Transmitters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lemberg, Jerry; Martelius, Mikko; Kosunen, Marko; Roverato, Enrico; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Ryynanen, Jussi

被引用次数: 4

2018

A common-gate common-source low noise amplifier based RF front end with selective input impedance matching for blocker-resilient receivers

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Ul Haq, Faizan; Ostman, Kim B.; Englund, Mikko; Stadius, Kari; Kosunen, Marko; Koli, Kimmo; Ryynanen, Jussi

被引用次数: 4

2018

A High-Speed DSP Engine for First-Order Hold Digital Phase Modulation in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Roverato, Enrico; Kosunen, Marko; Lemberg, Jerry; Martelius, Mikko; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Ryynanen, Jussi

被引用次数: 3

2018

Adaptive Nonlinear RF Cancellation for Improved Isolation in Simultaneous Transmit-Receive Systems

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kiayani, Adnan; Waheed, Muhammad Zeeshan; Anttila, Lauri; Abdelaziz, Mahmoud; Korpi, Dani; Syrjala, Ville; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 72

2018

Spatial Interpolation of Cyclostationary Test Statistics in Cognitive Radio Networks: Methods and Field Measurements

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY

Chaudhari, Sachin; Kosunen, Marko; Makinen, Semu; Ramanathan, Chandrasekaran; Oksanen, Jan; Laatta, Markus; Ryyanen, Jussi; Koivunen, Visa; Valkama, Mikko

被引用次数: 20

2018

Performance Analysis of Frequency-Reconfigurable Antenna Cluster With Integrated Radio Transceivers

IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS

Hannula, Jari-Matti; Kosunen, Marko; Lehtovuori, Anu; Rasilainen, Kimmo; Stadius, Kari; Ryyanen, Jussi; Viikari, Ville

被引用次数: 13

2018

A Systematic Design Method for Direct Delta-Sigma Receivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Englund, Mikko; Ul Haq, Faizan; Stadius, Kari; Kosunen, Marko; Ostman, Kim B.; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 2

2018

Reference Receiver Enhanced Digital Linearization of Wideband Direct-Conversion Receivers

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Marttila, Jaakko; Allen, Markus; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 18

2017

Modeling and Joint Mitigation of TX and RX Nonlinearity-Induced Receiver Desensitization

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kiayani, Adnan; Anttila, Lauri; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 16

2017

All-Digital LTE SAW-Less Transmitter With DSP-Based Programming of RX-Band Noise

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Roverato, Enrico; Kosunen, Marko; Cornelissens, Koen; Vatti, Sofia; Stynen, Paul; Bertrand, Kaoutar; Korhonen, Teuvo; Samsom, Hans; Vandenameele, Patrick; Ryyanen, Jussi

被引用次数: 16

2017

Digital Interpolating Phase Modulator for Wideband Outphasing Transmitters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lemberg, Jerry; Kosunen, Marko; Roverato, Enrico; Martelius, Mikko; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 26

2016

The effects of non-linearity in spectrum sensing receivers

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Turunen, Vesa; Kosunen, Marko; Kallioinen, Sami; Parssinen, Aarno; Ryyanen, Jussi

被引用次数: 0

2016

Performance Evaluation of Cyclostationary-Based Cooperative Sensing Using Field Measurements

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY

Chaudhari, Sachin; Kosunen, Marko; Makinen, Semu; Oksanen, Jan; Laatta, Markus; Ojaniemi, Jaakko; Koivunen, Visa; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 19

2016

RX-Band Noise Reduction in All-Digital Transmitters With Configurable Spectral Shaping of Quantization and Mismatch Errors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Roverato, Enrico; Kosunen, Marko; Lemberg, Jerry; Stadius, Kari; Ryyanen, Jussi

被引用次数: 8

2014

Survey and Analysis of Cyclostationary Signal Detector Implementations on FPGA

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kosunen, Marko; Turunen, Vesa; Kokkinen, Kari; Ryyanen, Jussi

被引用次数: 34

2013

Correlation-Based Detection of OFDM Signals in the Angular Domain

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY

Turunen, Vesa; Kosunen, Marko; Vaarakangas, Mikko; Ryyanen, Jussi

被引用次数: 20

2012

A Micropower $\Delta\Sigma$ -Based Interface ASIC for a Capacitive 3-Axis Micro-Accelerometer

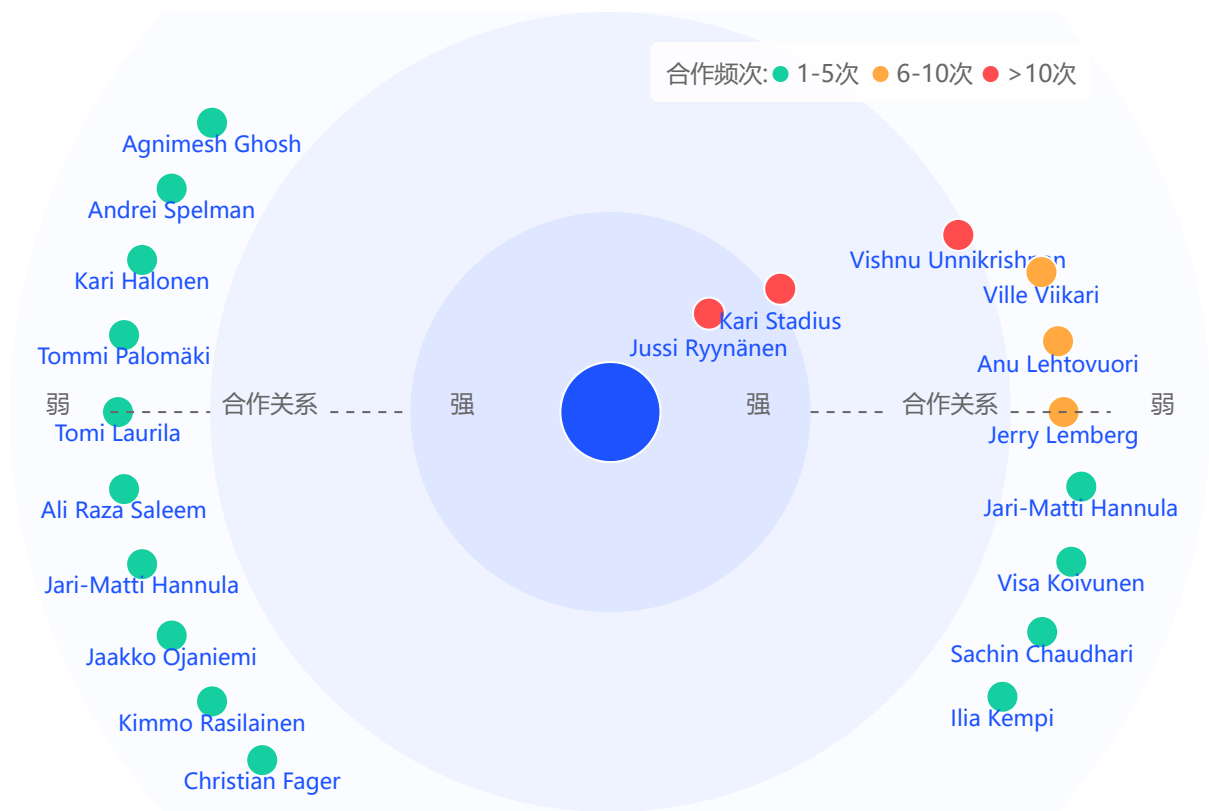
IEEE JOURNAL OF SOLID-STATE CIRCUITS

Paavola, Matti; Kamarainen, Mika; Laulainen, Erkka; Saukoski, Mikko; Koskinen, Lauri; Kosunen, Marko; Halonen, Kari A. I.

被引用次数: 66

2009

学术合作



Jussi Ryyänen

合作次数: 46

阿尔托大学

Kari Stadius

合作次数: 37

阿尔托大学

Vishnu Unnikrishnan

合作次数: 14

阿尔托大学

Ville Viikari

合作次数: 6

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Anu Lehtovuori

合作次数: 6

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